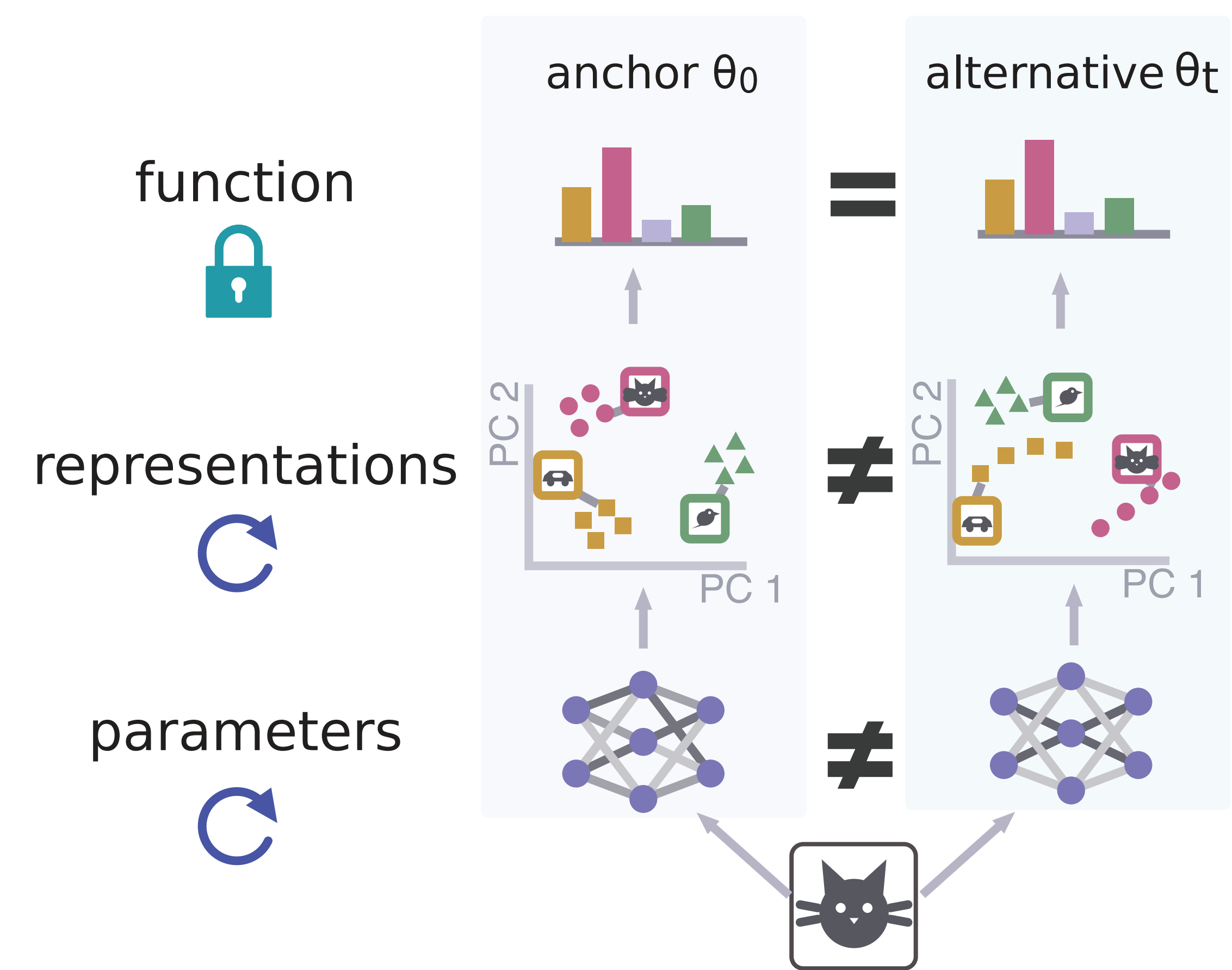


Traversing the Solution Space of Neural Networks with *Hessian Null Space Continuation*

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Function does not uniquely determine the mechanism



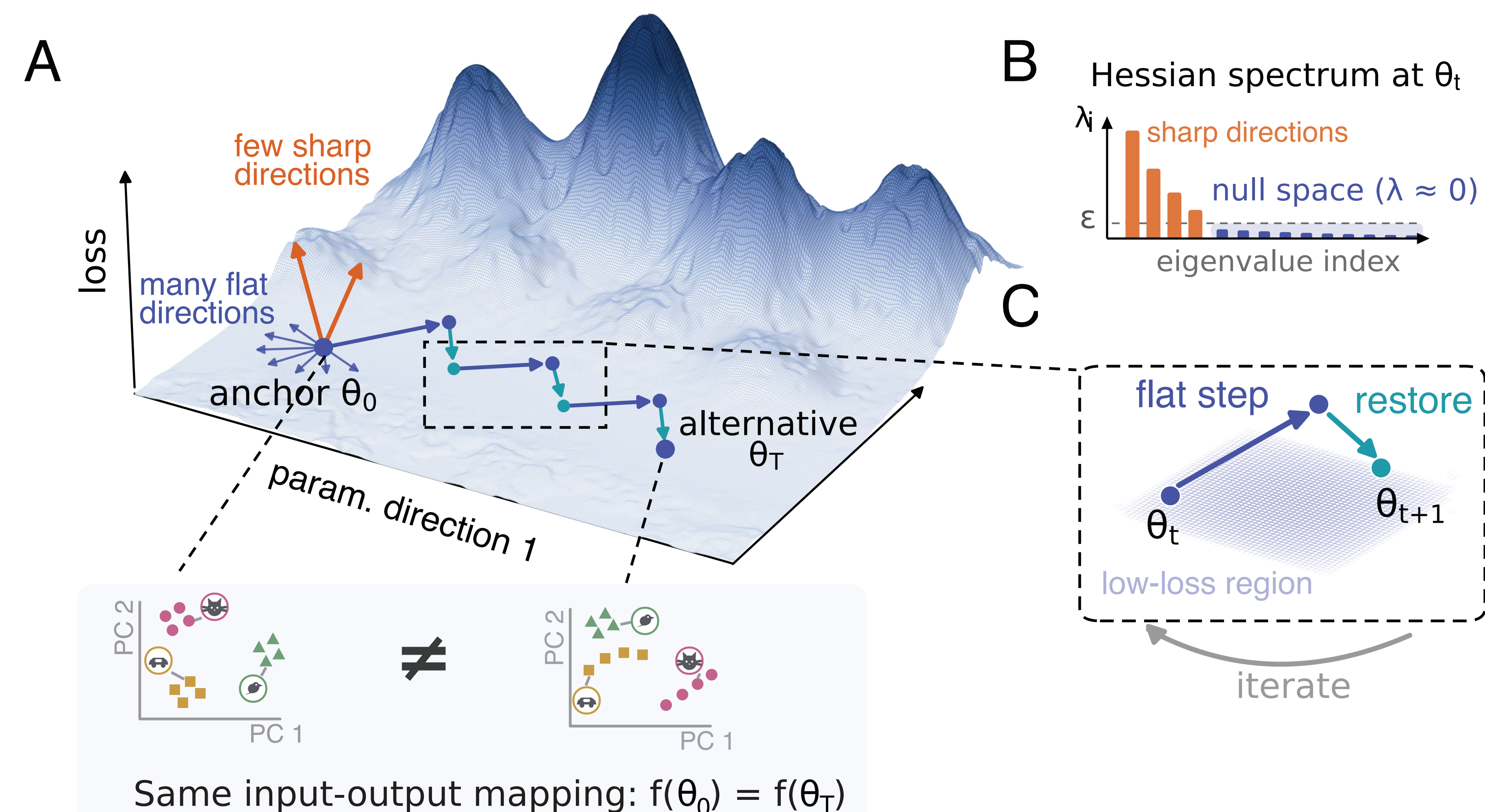
Recent work reveals non-unique solutions across seeds [1-3], dissociations between function and representation [4,5], and alternative solutions found by retraining [6,7].

Can we directly traverse the space of functionally equivalent solutions?

Hessian Null Space Continuation (HNC) traverses the solution space while preserving function

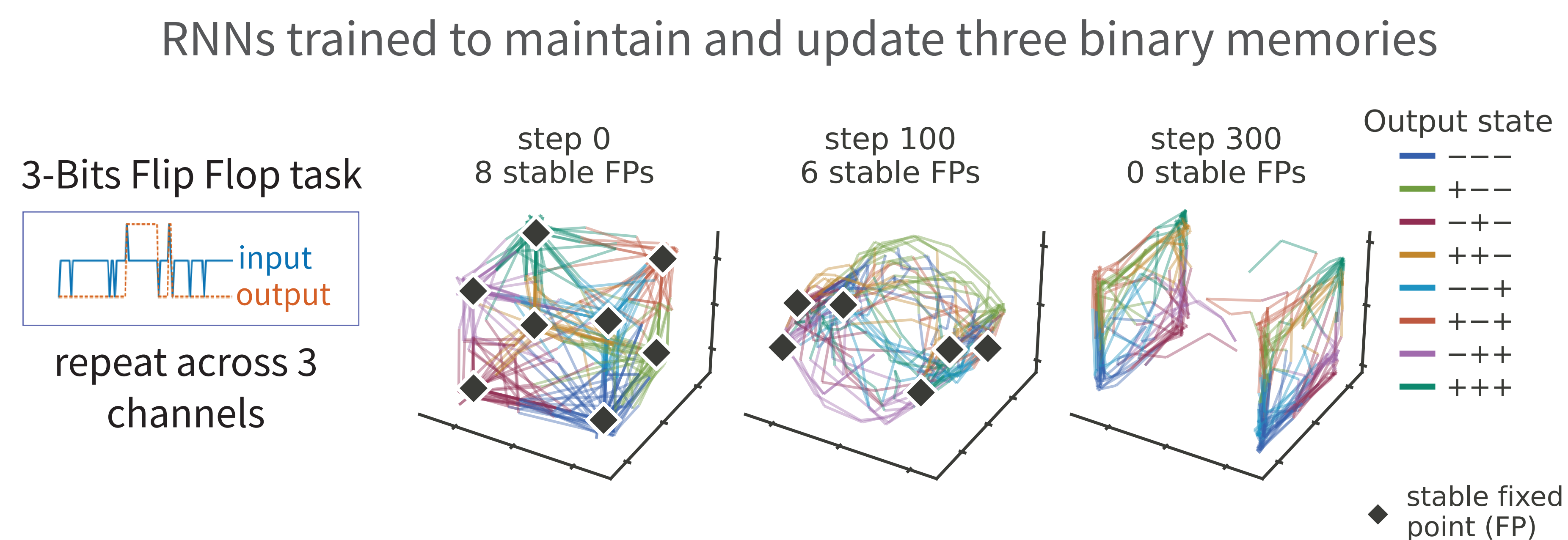
To preserve function, we define a function-matching loss

$$\mathcal{L}(\theta) = \frac{1}{2} \mathbb{E}_{x \sim \mathcal{D}} [\|f_{\theta}(x) - f_{\theta_0}(x)\|^2]$$

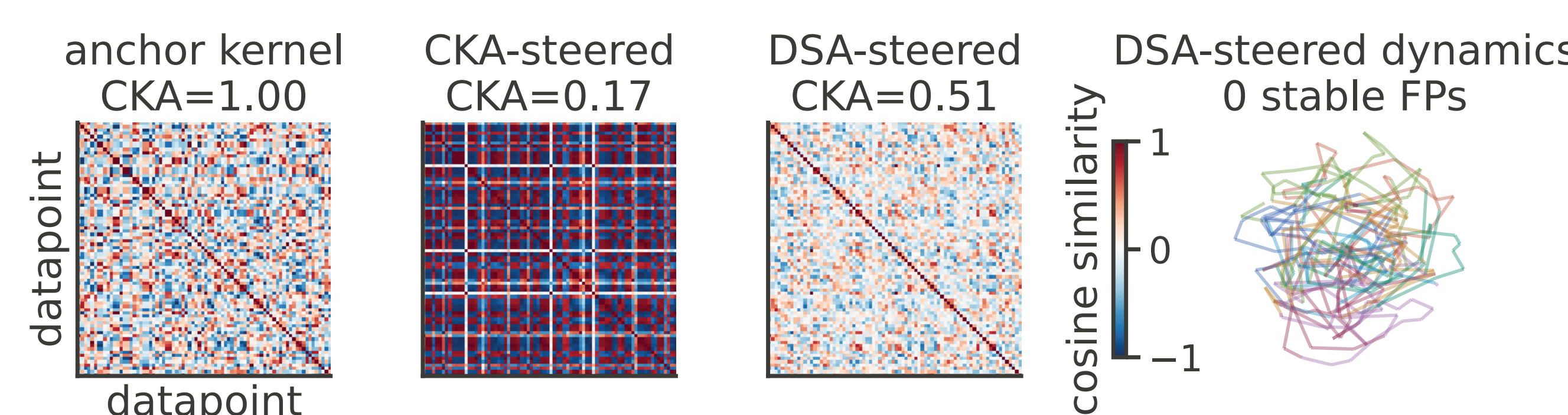


RNN

HNC drives RNNs to learn drastically different dynamics in a memory task

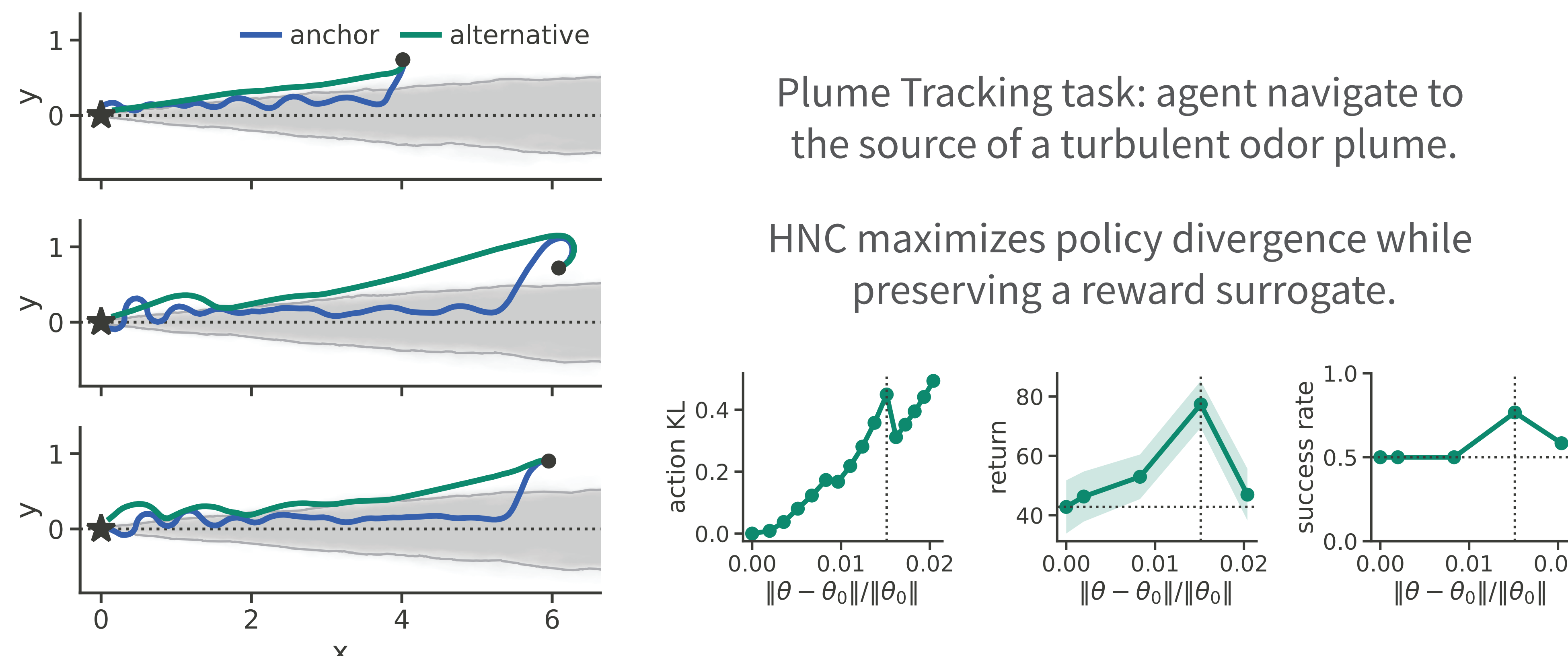


Steering to maximize representational and dynamical distance



RL

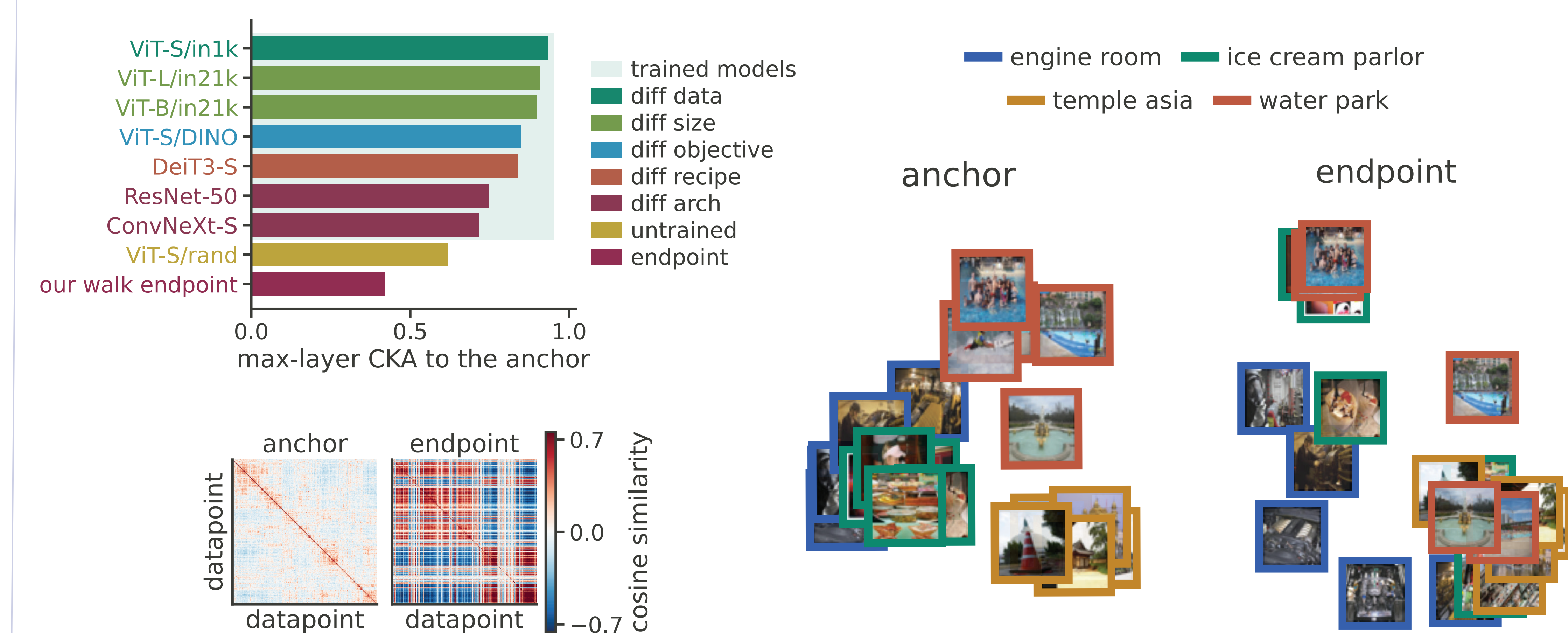
HNC finds behavioral distinct, high-reward policies in RL



ViT

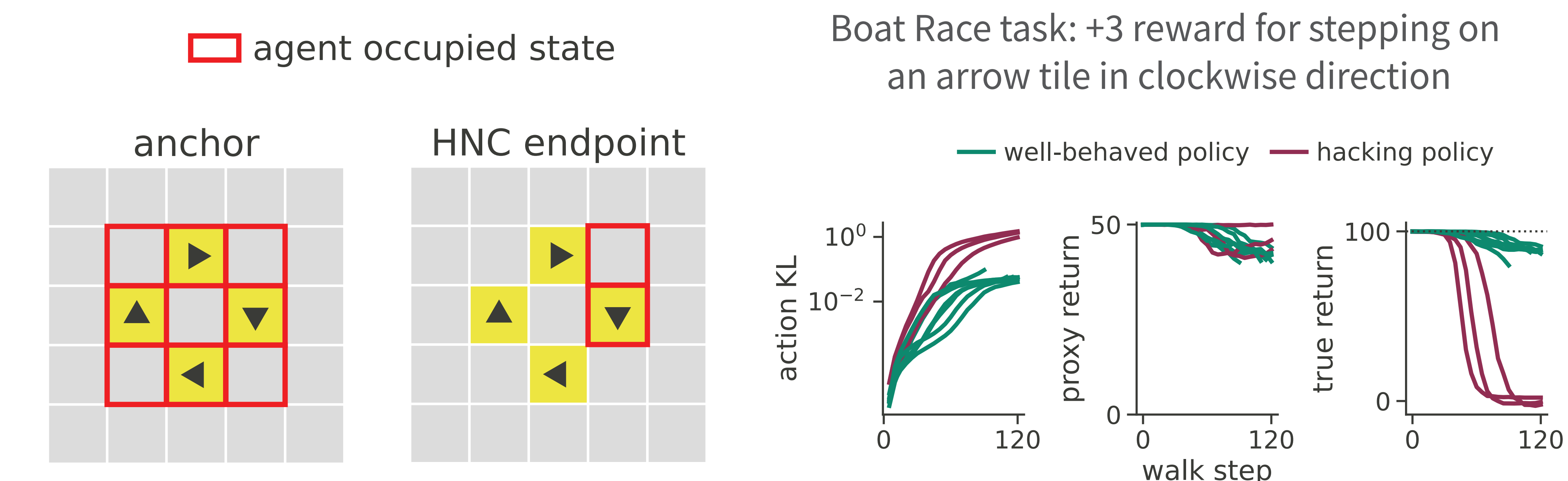
HNC reveals representational freedom underexplored by SGD in ViTs

ViT trained on ImageNet, steered to minimize Centered Kernel Alignment (CKA) from the anchor



RL

HNC exposes reward hacking solutions



Why does it matter?

Test whether an interpreted mechanism is actually general

Separate task constraints from optimizer bias

Model editing, fine-tuning, model merging